

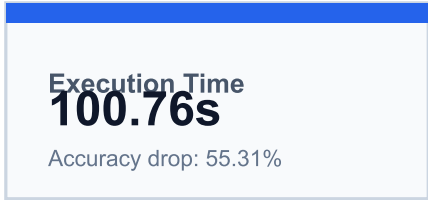
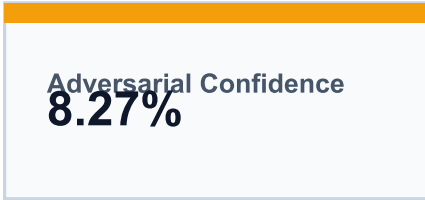
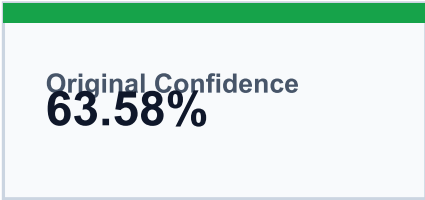
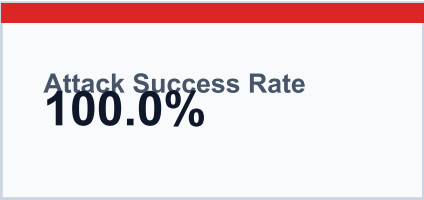
Threat Assessment Report

Executive summary for model robustness under adversarial attack

Model: facebook/convnext-tiny-224

Attack Type: DEEPFOOL

Images Evaluated: 9



Threat Level Assessment

HIGH RISK

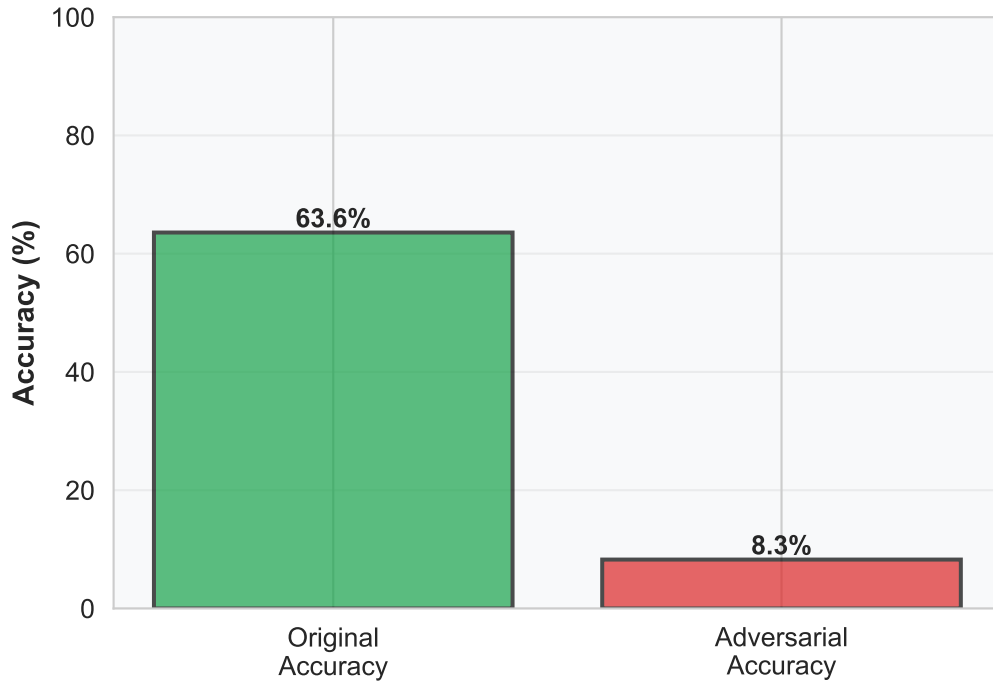
The DEEPFOOL attack altered model predictions on 100.0% of the evaluated images. Average confidence shifted from 63.58% to 8.27%, indicating the current robustness posture is high risk.

Assessment Narrative

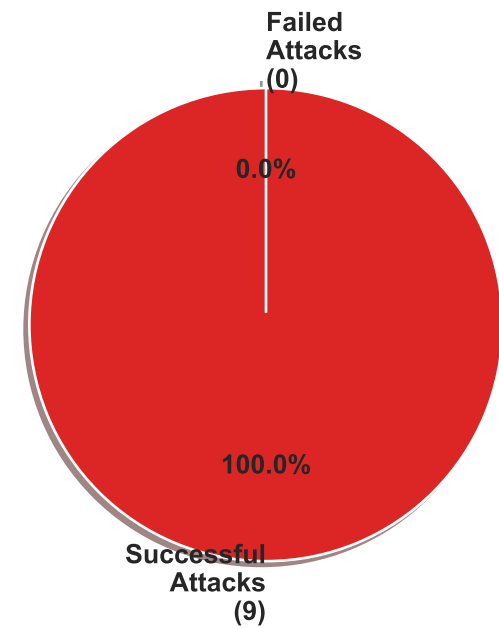
Successfully executed DEEPFOOL attack on model facebook/convnext-tiny-224 using 9 test images. Attack success rate: 100.0%. Average original accuracy: 63.58%, Average adversarial accuracy: 8.27%. The attack successfully fooled the model in 9 out of 9 cases.

Detailed Analysis

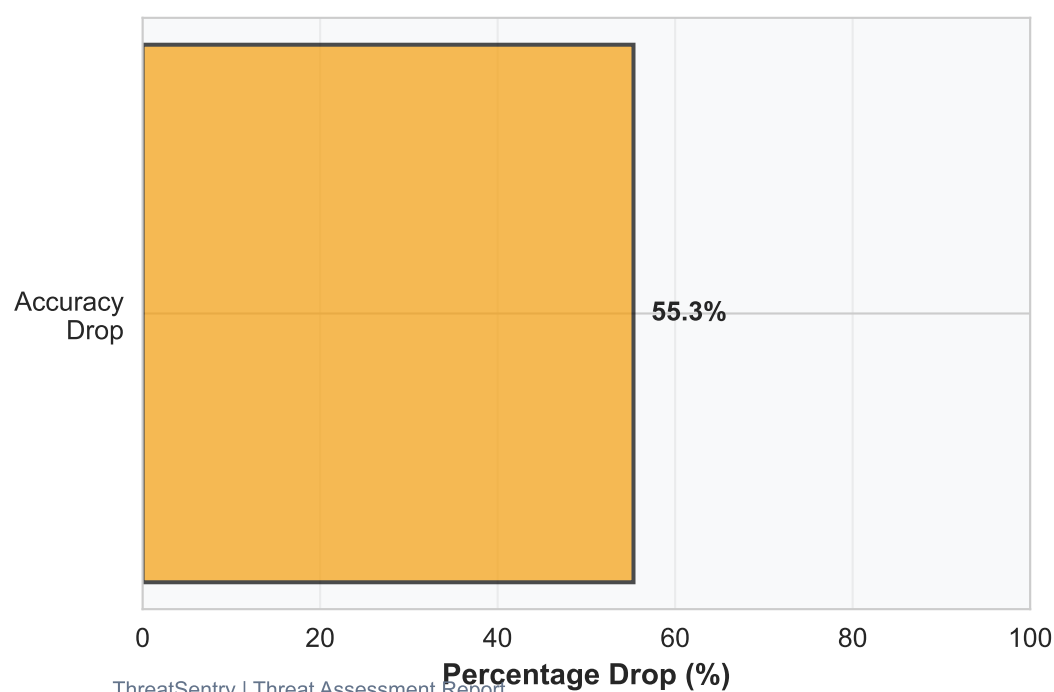
Accuracy Comparison



Attack Success Distribution



Model Robustness Impact



Confidence Comparison (Sample)

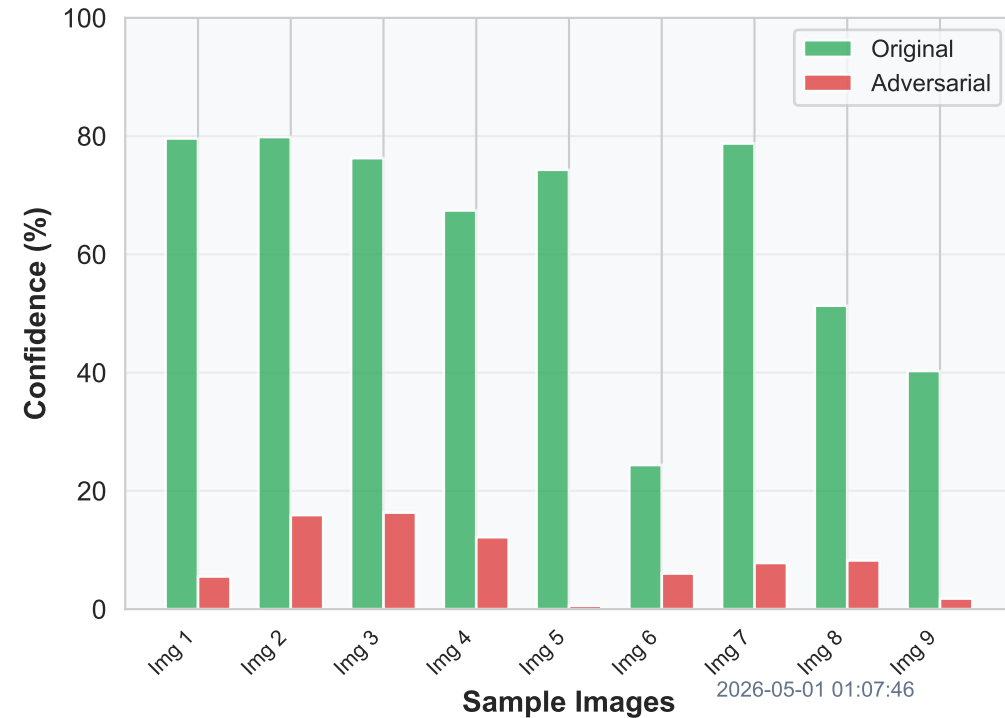


Image Evidence - Original vs Adversarial

Images 1-2 of 9

Original: coyote, prairie wolf, brush wolf, *Canis latrans*
(51.3%)



Adversarial: dhole, *Cuon alpinus*
(8.2%)
ATTACK SUCCESSFUL



Original: dhole, *Cuon alpinus*
(79.6%)



Adversarial: malinois
(5.5%)
ATTACK SUCCESSFUL



Image Evidence - Original vs Adversarial

Images 3-4 of 9

Original: Indian elephant, *Elephas maximus*
(67.4%)



Adversarial: tusker
(12.2%)
ATTACK SUCCESSFUL



Original: tiger, *Panthera tigris*
(79.8%)



Adversarial: rule, ruler
(15.9%)
ATTACK SUCCESSFUL



Image Evidence - Original vs Adversarial

Images 5-6 of 9

Original: ox
(24.4%)



Original: gibbon, Hylobates lar
(78.8%)



Adversarial: water buffalo, water ox, Asiatic buffalo, Bubalus bubalis
(6.0%)
ATTACK SUCCESSFUL



Adversarial: howler monkey, howler
(7.8%)
ATTACK SUCCESSFUL

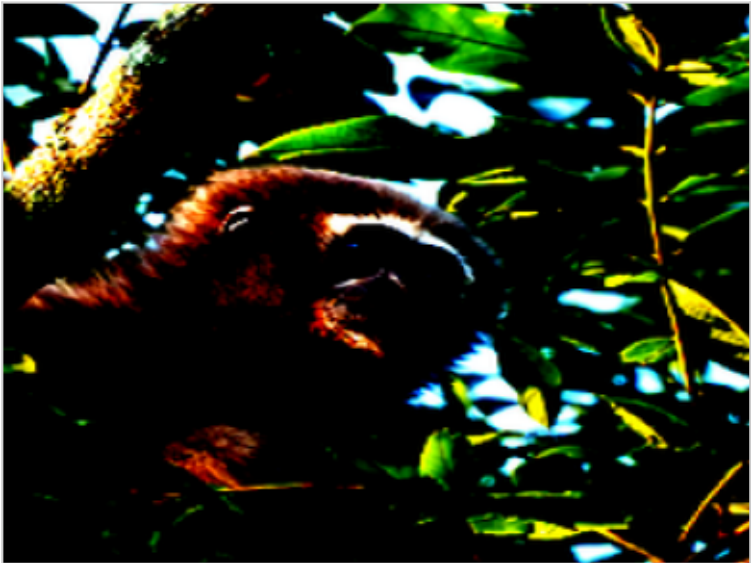


Image Evidence - Original vs Adversarial

Images 7-8 of 9

Original: capuchin, ringtail, Cebus capucinus
(40.3%)



Adversarial: bee eater
(1.8%)
ATTACK SUCCESSFUL



Original: water buffalo, water ox, Asiatic buffalo, Bubalus bubalis
(74.3%)



Adversarial: American coot, marsh hen, water hen, Fulica americana
(0.6%)
ATTACK SUCCESSFUL

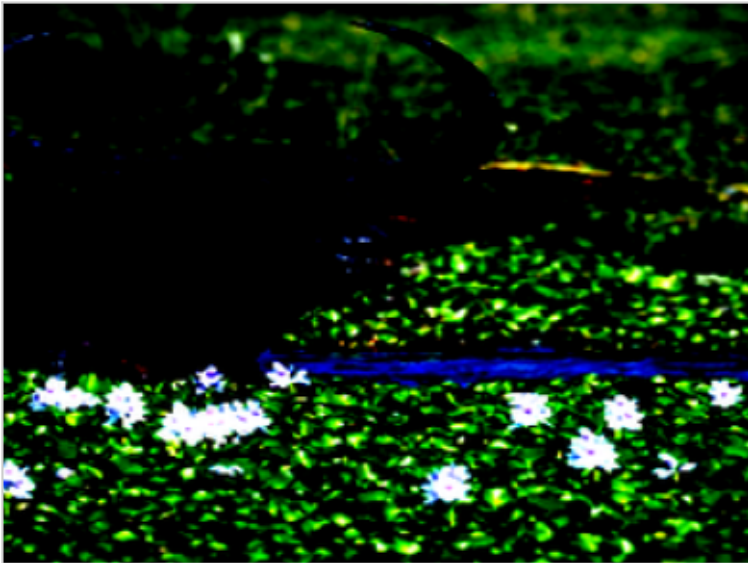


Image Evidence - Original vs Adversarial

Images 9-9 of 9

Original: ibex, Capra ibex
(76.3%)

Adversarial: gazelle
(16.3%)
ATTACK SUCCESSFUL



Recommendations

Security actions based on observed threat exposure

Assessment Summary

Successfully executed DEEPFOOL attack on model facebook/convnext-tiny-224 using 9 test images. Attack success rate: 100.0%. Average original accuracy: 63.58%, Average adversarial accuracy: 8.27%. The attack successfully fooled the model in 9 out of 9 cases.

Security Recommendations

1. Implement Adversarial Training

- Retrain your model with adversarial examples to improve robustness
- Use techniques like FGSM, PGD during training phase

2. Add Input Validation & Preprocessing

- Implement input sanitization and anomaly detection
- Use defensive distillation or feature squeezing

3. Deploy Ensemble Methods

- Use multiple models with different architectures
- Implement voting mechanisms for predictions

4. Continuous Monitoring

- Set up real-time performance monitoring
- Detect and alert on unusual prediction patterns

5. Regular Security Audits

- Conduct periodic threat assessments
- Stay updated with latest attack techniques